Web Dev Mastery

# 🌟 CSS Flexbox Masterclass

This masterclass covers **every Flexbox property** with detailed explanations, use-cases, and HTML + CSS code snippets.

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## 🔰 Getting Started

### What is Flexbox?

Flexbox is a CSS layout model designed to arrange items in a **row or column**, distribute space dynamically, and align content easily.

To enable Flexbox:

```
.parent {
  display: flex;
}
```

HTML Example:

```
<div class="parent">
  <div class="child">Item 1</div>
  <div class="child">Item 2</div>
</div>
```

---

# 🧩 Flex Container Properties (Apply to parent)

## 1. `display: flex;`

- Turns the element into a flex container.
- Required to use any other Flexbox features.

## 2. `flex-direction`

- Defines the direction of the main axis (i.e., the direction the children flow).

**Values:**

- `row` (default): Left to right.
- `row-reverse`: Right to left.
- `column`: Top to bottom.
- `column-reverse`: Bottom to top.

```
.parent {
  display: flex;
  flex-direction: row;
}
```

## 3. `flex-wrap`

- Determines whether flex items should wrap or stay on one line.

**Values:**

- `nowrap` (default): All items in one line.
- `wrap`: Items wrap to next line.

- `wrap-reverse`: Wraps in reverse direction.

```
.parent {
  display: flex;
  flex-wrap: wrap;
}
```

## 4. `justify-content`

- Aligns items **horizontally** along the main axis.

**Values:**

- `flex-start`: Align to left (default for row).
- `flex-end`: Align to right.
- `center`: Center horizontally.
- `space-between`: Equal space between.
- `space-around`: Equal space around.
- `space-evenly`: Equal space between and around.

```
.parent {
  display: flex;
  justify-content: space-between;
}
```

## 5. `align-items`

- Aligns items **vertically** along the cross axis.

**Values:**

- `stretch` (default): Items stretch to fill.

- `flex-start`: Top-aligned.
- `flex-end`: Bottom-aligned.
- `center`: Vertically centered.
- `baseline`: Align based on text baseline.

```
.parent {
  display: flex;
  align-items: center;
}
```

## 6. `align-content`

- Aligns multiple rows (if items wrap).

**Values:**

- Same as `justify-content`: `flex-start`, `center`, `space-between`, etc.

```
.parent {
  display: flex;
  flex-wrap: wrap;
  align-content: space-around;
}
```

---

# 🎯 Flex Item Properties (Apply to children)

## ✅ 1. `order` – Controls visual order

### 🔹 HTML

```
<h2>1. Flex Item Property: order</h2>
<div class="container">
  <div class="item">First (Order 2)</div>
  <div class="item">Second (Order 1)</div>
  <div class="item">Third (Order 0)</div>
</div>
```

### 🔹 CSS

```
.container {
  display: flex;
  gap: 10px;
}
.item {
  padding: 10px 20px;
  background-color: #009688;
  color: white;
  border-radius: 5px;
}

/* Applying order */
.item:nth-child(1) {
  order: 2;
}
.item:nth-child(2) {
  order: 1;
}
.item:nth-child(3) {
  order: 0;
}
```

---

## ✅ 2. `flex-grow` – Defines how items grow

### 🔹 HTML

```
<h2>2. Flex Item Property: flex-grow</h2>
<div class="container">
  <div class="item">Grow 1</div>
  <div class="item">Grow 2</div>
</div>
```

### 🔹 CSS

```
.container {
  display: flex;
  gap: 10px;
}
.item {
  flex-grow: 1;
  padding: 10px;
  background-color: #3f51b5;
  color: white;
  text-align: center;
}
```

---

## ✅ 3. `flex-shrink` – Defines how items shrink

### 🔹 HTML

```
<h2>3. Flex Item Property: flex-shrink</h2>
<div class="container">
  <div class="item">Shrink 1</div>
  <div class="item">Shrink 2</div>
</div>
```

### 🔹 CSS

```
.container {
  display: flex;
  gap: 10px;
  width: 200px;
}
.item {
  flex-shrink: 1;
  min-width: 150px;
  padding: 10px;
  background-color: #ff9800;
  color: white;
  text-align: center;
}
```

---

## ✅ 4. `flex-basis` – Sets initial item size

### 🔹 HTML

```
<h2>4. Flex Item Property: flex-basis</h2>
<div class="container">
  <div class="item">200px</div>
  <div class="item">200px</div>
</div>
```

### 🔹 CSS

```
.container {
  display: flex;
  gap: 10px;
}
.item {
  flex-basis: 200px;
  padding: 10px;
  background-color: #4caf50;
  color: white;
```

```
  text-align: center;
}
```

---

## ✅ 5. `flex` – Shorthand for grow, shrink, basis

### 🔹 HTML

```
<h2>5. Flex Item Property: flex (shorthand)</h2>
<div class="container">
  <div class="item">flex: 1 1 200px</div>
  <div class="item">flex: 1 1 200px</div>
</div>
```

### 🔹 CSS

```
.container {
  display: flex;
  gap: 10px;
}
.item {
  flex: 1 1 200px; /* grow shrink basis */
  padding: 10px;
  background-color: #9c27b0;
  color: white;
  text-align: center;
}
```

---

## ✅ 6. `align-self` – Override align-items

### 🔹 HTML

```
<h2>6. Flex Item Property: align-self</h2>
<div class="container">
  <div class="item">Top (default)</div>
  <div class="item">Bottom (align-self)</div>
</div>
```

### 🔹 CSS

```
.container {
  display: flex;
  align-items: flex-start;
  height: 150px;
  gap: 10px;
  background: #e0f7fa;
}
.item {
  background-color: #f44336;
  color: white;
  padding: 10px;
  border-radius: 5px;
}
.item:last-child {
  align-self: flex-end;
}
```

---

## ✅ Summary Table

| Property | Applied To | Description |
|---|---|---|
| `display` | Parent | Turns on Flexbox |
| `flex-direction` | Parent | Row or column layout |
| `flex-wrap` | Parent | Wraps items if needed |
| `justify-content` | Parent | Aligns items on main axis |
| `align-items` | Parent | Aligns on cross axis |
| `align-content` | Parent | Aligns multiple lines |

| | | |
|---|---|---|
| `order` | Child | Reorders items |
| `flex-grow` | Child | Defines grow factor |
| `flex-shrink` | Child | Shrink factor |
| `flex-basis` | Child | Initial size |
| `flex` | Child | Shorthand for grow/shrink/basis |
| `align-self` | Child | Individual alignment |

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## 📚 Practice Ideas

- Build a responsive navbar
- Create a pricing table with equal-width cards
- Make a responsive photo gallery
- Center a modal box both horizontally and vertically
- Build a dashboard layout with sidebar + content

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